

Wave Generator

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In the world of hobbyists, the function generator ranks pretty high in the list of essential items for the lab bench. However, commercial units can be very expensive. This makes function generators a prime candidate for home construction.

Almost all function generator designs are based on the industry-standard IC L8038 chip and have a fair degree of similarity. The design presented here is based on a slightly different principle for a number of reasons.

First, it is more interesting to approach the problem with some fresh thought rather than repeating the earlier concept. Of course, one should try to avoid reinventing the wheel. Second, for this generator, it was desired

to provide control for both frequency and amplitude. Third, the technique used for sine-wave generation has certain advantages. In this design, triangle wave is passed through a low-pass filter. Although the filter used is fairly simple and does not eliminate distortion completely, it produces a less spiky output that is better suited for the testing of audio amplifiers.

Obviously, in order to achieve a constant output level, the filter cut-off frequency must be made to track the signal frequency. This has been achieved by making both the oscillator and the filter voltage-controlled.

Circuit description

Fig. 1 shows the wave generator circuit. It is built around two LM13700 dual operational transconductance amplifiers (IC1 and IC2), high-per-

formance op-amp NE5534 (IC3), dual JFET input op-amp TL082 (IC4) and a few discrete components.

The power supply circuit (shown in Fig. 2) is built around regulator ICs 7812 (IC5) and 7912 (IC6) and a few discrete components. The pin configurations of regulator ICs 7812 and 7912 are shown in Fig. 3.

Five-way rotary switch S2 is used as the range selector, while three-way rotary switch S3 is the mode selector. Positions 1, 2, 3, 4 and 5 of range-selection switch S2 give frequency ranges of 1-10 Hz, 10-100 Hz, 0.1-1 kHz, 1-10 kHz and 10-75 kHz, respectively. Positions 1, 2 and 3 of mode-selection switch S3 give square wave, sine wave and triangular wave, respectively.

In the wave generator circuit, the oscillator is formed by op-amps A1

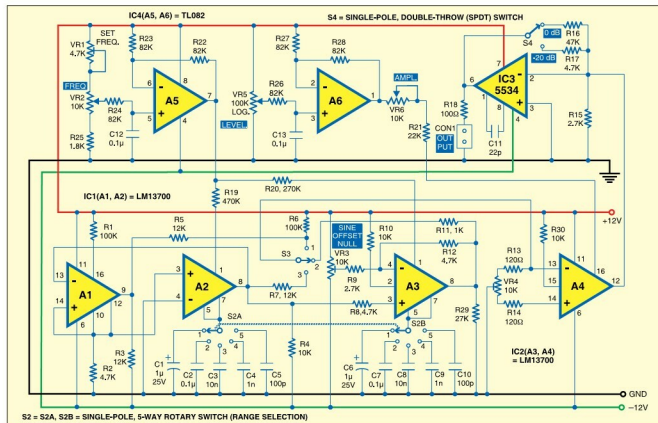


Fig. 1: Circuit for the wave generator